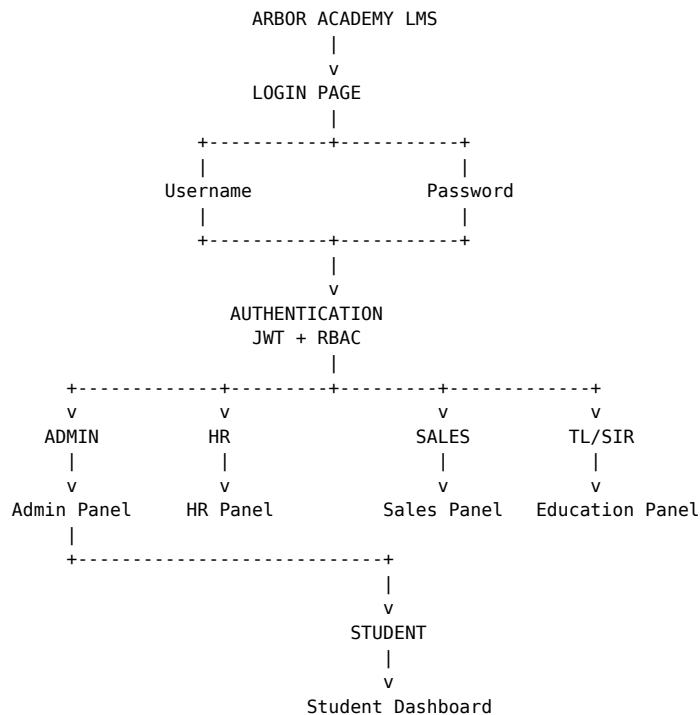


Arbor Academy LMS Portal

Complete project blueprint: page flow, role permissions, course/batch workflow, React + Vite frontend, Node.js + Express backend, database, Google services and AWS deployment.

Recommended stack: React + Vite, React Router, Axios, Context API, Tailwind CSS, Node.js, Express.js, MongoDB + Mongoose, JWT/RBAC, Google Drive API, Google Sheets API, Email Service, AWS S3, EC2 and CloudFront.

1. COMPLETE ARBOR ACADEMY LMS FLOW



2. LOGIN → ROLE-BASED PAGE FLOW

```
LOGIN
|
v
Enter Username / Email + Password
|
v
Backend validates credentials
|
v
JWT token generated
|
v
Role checked
|
+-- ADMIN    -> /admin/dashboard
|
+-- HR       -> /hr/dashboard
|
+-- SALES    -> /sales/dashboard
|
+-- TL       -> /tl/dashboard
|
+-- TRAINER  -> /trainer/dashboard
|
+-- STUDENT  -> /student/dashboard
|
+-- Other/Invalid -> /unauthorized
```

3. ADMIN PAGE STRUCTURE

```
ADMIN LOGIN
|
v
ADMIN DASHBOARD
|
+-- Dashboard
+-- User Management
|   +-- Admin Users
|   +-- HR Users
|   +-- Sales Users
|   +-- TL Users
|   +-- Trainer Users
|   +-- Students
|
+-- Department Management
|   +-- HR
|   +-- Sales
|   +-- Education
|
+-- Course Management
|   +-- Full Stack Development
|   +-- Data Analytics
|   +-- Data Science
|   +-- AWS DevOps
|   +-- UI/UX
|   +-- Cyber Security
|   +-- AI/ML
|
+-- Batch Management
+-- Trainer Management
+-- Student Management
+-- Reports
+-- Roles / Permissions
+-- System Settings
```

4. SALES DEPARTMENT FLOW

```
SALES LOGIN
|
v
SALES DASHBOARD
|
+-- Dashboard
+-- New Admission
|   +-- Student Name
|   +-- Mobile
|   +-- Email
|   +-- Course
|   +-- Batch
|   +-- TL
|   +-- Admission Date
|   +-- Fees
|
+-- Fee Management
|   +-- Paid
|   +-- Pending
|   +-- Partial
|
+-- Upload Fee Receipt
+-- Admission Records
+-- Sales Reports
```

Example:

```
Prachi (Sales)
-> New Admission
-> Sayali Potle
-> AWS DevOps
-> August 2026 Batch
-> TL: Kunal Sir
-> Fee received by GP
-> Receipt uploaded
-> Student ID generated
-> Initial password generated
-> Course + Batch + TL + Trainer assigned
```

5. STUDENT ID + INITIAL PASSWORD FLOW

```
Admission + Fee Receipt
|
v
Backend creates student record
|
v
Generate unique Student ID
|
v
Generate secure initial PIN/password
|
v
Example:
Student: Sayali Potle
Student ID: 41548745
Generated PIN: 4827
Initial Password: Sayali@4827
|
v
Student can login
```

Recommended security:
Initial password -> first login -> force password change.
Do not store plain passwords in the database; store a secure hash.

6. TL DASHBOARD

```
TL LOGIN
|
v
TL DASHBOARD
|
+-- Dashboard
+-- My Batches
|   +-- AWS DevOps - Aug 2026
|   +-- AWS DevOps - Sep 2026
|   +-- Full Stack - Aug 2026
|
+-- Student Management
|   +-- Add Student to Batch
|   +-- Remove / Transfer Student
|   +-- Student Details
|
+-- Trainer Management
|   +-- Assign Trainer
|   +-- Change Trainer
|   +-- Trainer Performance
|
+-- Course Progress
+-- Attendance
+-- Assignment Progress
+-- Exam Progress
+-- Student Performance
+-- Batch Performance
+-- Trainer Performance
+-- Ready for Interview
+-- Reports
```

7. TL RESPONSIBILITIES

```
TL Kunal Sir
|
+-- Assign Student -> Batch
+-- Assign Trainer -> Batch
+-- Change Trainer
+-- Transfer Student
+-- View Attendance
+-- View Course Progress
+-- View Assignments
+-- View Exams
+-- View Student Performance
+-- View Trainer Performance
+-- Download Batch Reports
+-- Verify Interview Readiness
```

+-- Move Eligible Student -> HR

8. TRAINER / SIR DASHBOARD

SIR LOGIN

|

v

TRAINER DASHBOARD

|

- +-- My Courses
- +-- My Batches
- +-- My Students
- +-- Attendance
- +-- Course Topics
- +-- Student Progress
- +-- Assignments
- +-- Exams
- +-- Performance
- +-- Reports
- +-- Interview Recommendation

Trainer should not control:

- Admission
- Fee status
- Receipt verification
- User roles
- HR placement decisions

9. STUDENT PROGRESS RECORD

Student: Soham Patil

|

- +-- Attendance
 - | +-- Total Classes: 40
 - | +-- Present: 37
 - | +-- Absent: 3
 - | +-- Attendance: 92.5%

|

- +-- Assignments
 - | +-- Given: 15
 - | +-- Submitted: 13
 - | +-- Pending: 2
 - | +-- Score: 86%

|

- +-- Exams
 - | +-- Given: 8
 - | +-- Passed: 7
 - | +-- Failed: 1
 - | +-- Average: 82%

|

- +-- Course Progress: 68%

|

- +-- Interview Readiness
 - +-- Technical: 78%
 - +-- Attendance: 92%
 - +-- Assignment: 86%
 - +-- Exam: 82%
 - +-- Ready: YES

10. COURSE → TOPIC → PROGRESS

AWS DEVOPS

|

- +-- Linux
 - | +-- Linux Basics [#####] 100%
 - | +-- Commands [#####] 100%
 - | +-- Shell Scripting [#####] 70%

|

- +-- AWS
 - | +-- IAM [#####] 100%
 - | +-- EC2 [#####] 100%
 - | +-- S3 [#####] 80%
 - | +-- VPC [#####] 50%
 - | +-- Lambda [#####] 30%
 - | +-- CloudFront [#####] 0%

```

|
+-- Docker
+-- Kubernetes
+-- CI/CD
|
+-- Overall Course Progress

```

11. WHO CAN SEE PROGRESS?

STUDENT

-> Own progress only

TRAINER / SIR

-> Assigned students / assigned batches

TL

-> All students in TL-managed batches
-> Trainer performance
-> Batch performance

ADMIN

-> All students, batches, trainers and reports

HR

-> Students moved to interview / placement workflow

12. HR DASHBOARD + INTERVIEW FLOW

HR LOGIN

```

|
v
HR DASHBOARD
|
+-- Dashboard
+-- Ready for Interview
+-- Interview Schedule
+-- Technical Round
+-- HR Round
+-- Interview Results
|   +-- Selected
|   +-- Rejected
|   +-- On Hold
|
+-- Placement
+-- HR Reports

```

Readiness workflow:

```

Student completes required topics
|
v
Required attendance achieved
|
v
Assignments completed
|
v
Required exams passed
|
v
Trainer recommends
|
v
TL verifies
|
v
READY FOR INTERVIEW
|
v
HR schedules interview
|
v
Selected / Rejected / On Hold
|
v
Placement

```

13. STUDENT DASHBOARD

```
STUDENT LOGIN
|
v
STUDENT DASHBOARD
|
+-- Home
+-- My Profile
+-- My Course
+-- My Batch
+-- Course Topics
+-- Lecture Videos
+-- My Attendance
+-- Assignments
+-- Exams
+-- My Performance
+-- Progress Report
+-- Certificates
+-- Interview Status
+-- Placement Status
```

14. COMPLETE STUDENT JOURNEY

```
SALES
-> Admission Form
-> Fees Received
-> Receipt Uploaded
-> Student ID Generated
-> Course Assigned
-> Batch Assigned
-> TL Assigned
-> Trainer Assigned
-> Student Login
-> Attend Classes
-> Watch Lectures
-> Complete Topics
-> Submit Assignments
-> Give Exams
-> Trainer Tracks Progress
-> TL Reviews Student
-> Ready for Interview?
    |
    +-- NO -> Continue Training
    |
    +-- YES -> HR
        -> Interview
        -> Selected / Rejected / On Hold
        -> Placement
```

15. REACT + VITE FRONTEND FOLDER STRUCTURE

```
arbor-lms/
|
+-- frontend/
|   |
|   +-- public/
|       +-- logo.png
|       +-- favicon.ico
|   |
|   +-- src/
|       |
|       +-- assets/
|           |
|           +-- images/
|           +-- icons/
|       |
|       +-- components/
|           |
|           +-- common/
|           +-- dashboard/
|           +-- students/
|           +-- courses/
|           +-- batches/
|           +-- attendance/
|           +-- assignments/
|           +-- exams/
|           +-- reports/
```

```

|
| +-- pages/
| | +-- auth/
| | +-- admin/
| | +-- sales/
| | +-- tl/
| | +-- trainer/
| | +-- hr/
| | +-- student/
| |
| +-- context/
| | +-- AuthContext.jsx
| | +-- UserContext.jsx
| |
| +-- hooks/
| | +-- useAuth.js
| | +-- usePermission.js
| |
| +-- routes/
| | +-- AppRoutes.jsx
| | +-- ProtectedRoute.jsx
| | +-- RoleRoute.jsx
| |
| +-- services/
| | +-- api.js
| | +-- authService.js
| | +-- studentService.js
| | +-- courseService.js
| | +-- batchService.js
| | +-- reportService.js
| |
| +-- utils/
| +-- App.jsx
| +-- main.jsx
| +-- index.css
|
| +-- .env
| +-- package.json
| +-- vite.config.js

```

16. NODE.JS + EXPRESS BACKEND FOLDER STRUCTURE

```

backend/
|
+-- src/
| |
| | +-- config/
| | | +-- db.js
| | | +-- googleDrive.js
| | | +-- googleSheets.js
| |
| | +-- controllers/
| | | +-- authController.js
| | | +-- adminController.js
| | | +-- salesController.js
| | | +-- tlController.js
| | | +-- trainerController.js
| | | +-- studentController.js
| | | +-- hrController.js
| | | +-- courseController.js
| | | +-- batchController.js
| | | +-- attendanceController.js
| | | +-- assignmentController.js
| | | +-- examController.js
| | | +-- progressController.js
| | | +-- reportController.js
| |
| | +-- models/
| | | +-- User.js
| | | +-- Student.js
| | | +-- Course.js
| | | +-- Topic.js
| | | +-- Batch.js
| | | +-- Trainer.js
| | | +-- Attendance.js
| | | +-- Assignment.js
| | | +-- Exam.js
| | | +-- Progress.js
| | | +-- Admission.js

```

```

| | +-- FeeReceipt.js
| | +-- Interview.js
| | +-- Placement.js
| |
| +-- routes/
| | +-- authRoutes.js
| | +-- adminRoutes.js
| | +-- salesRoutes.js
| | +-- tlRoutes.js
| | +-- trainerRoutes.js
| | +-- studentRoutes.js
| | +-- hrRoutes.js
| | +-- courseRoutes.js
| | +-- batchRoutes.js
| | +-- attendanceRoutes.js
| | +-- assignmentRoutes.js
| | +-- examRoutes.js
| | +-- progressRoutes.js
| | +-- reportRoutes.js
| |
| +-- middleware/
| | +-- authMiddleware.js
| | +-- roleMiddleware.js
| | +-- uploadMiddleware.js
| | +-- errorMiddleware.js
| |
| +-- services/
| | +-- studentIdService.js
| | +-- passwordService.js
| | +-- googleDriveService.js
| | +-- googleSheetService.js
| | +-- emailService.js
| | +-- reportService.js
| |
| +-- utils/
| | +-- generateStudentId.js
| | +-- generatePassword.js
| | +-- calculateProgress.js
| |
| +-- app.js
| +-- server.js
|
+-- .env
+-- package.json
+-- README.md

```

17. DATABASE STRUCTURE

USER

```

|
+-- ADMIN
+-- HR
+-- SALES
+-- TL
+-- TRAINER
+-- STUDENT

```

COURSE

```

|
+-- BATCH
|
|   +-- TL
|   +-- TRAINER
|   +-- STUDENTS
|       |
|       +-- Attendance
|       +-- Progress
|       +-- Assignments
|       +-- Exams
|       +-- Performance
|       +-- Interview
|       +-- Placement

```

18. STUDENT RECORD MODEL

Student

```

|

```



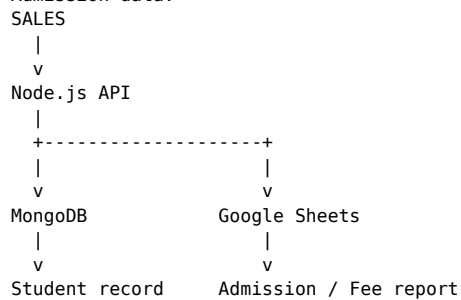
```

+-- studentId
+-- name
+-- email
+-- mobile
+-- courseId
+-- batchId
+-- tId
+-- trainerId
+-- admissionId
+-- feeStatus
+-- receiptUrl
+-- attendance
+-- assignments
+-- exams
+-- progress
+-- interviewStatus
+-- placementStatus

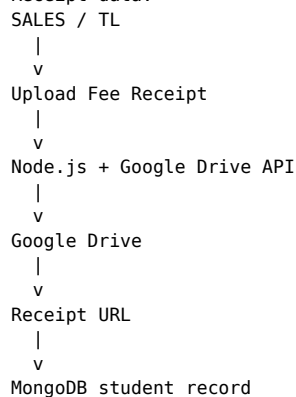
```

19. GOOGLE SHEETS + GOOGLE DRIVE FLOW

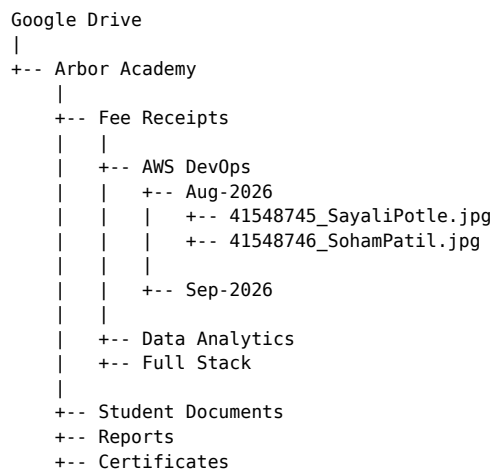
Admission data:



Receipt data:



20. GOOGLE DRIVE FOLDER STRUCTURE



21. GOOGLE SHEETS STRUCTURE

Arbor Academy LMS
|
+-- Students
+-- Admissions
+-- Fees
+-- Attendance
+-- Assignments
+-- Exams
+-- Progress
+-- Interviews
+-- Placements
+-- Trainer Performance
+-- Batch Reports

22. ROLE PERMISSION STRUCTURE

ADMIN
-> Everything

HR
-> Interview + Placement + HR Student Records

SALES
-> Admission + Fees + Receipt + Student Creation

TL
-> Batch + Student + Trainer + Progress + Reports

TRAINER / SIR
-> Attendance + Topics + Assignment + Exam + Performance

STUDENT
-> Own Course + Attendance + Assignment + Exam + Progress

23. RECOMMENDED PAGE COUNT

AUTH
+-- Login
+-- Forgot Password
+-- Unauthorized
= 3 major pages

ADMIN
+-- Dashboard
+-- Users
+-- Departments
+-- Courses
+-- Batches
+-- Trainers
+-- Students
+-- Reports
+-- Settings
= 9

SALES
+-- Dashboard
+-- Admissions
+-- New Admission
+-- Fee Management
+-- Receipt Upload
+-- Reports
= 6

TL
+-- Dashboard
+-- My Batches
+-- Students
+-- Trainers
+-- Student Progress
+-- Batch Progress
+-- Reports
+-- Interview Ready
= 8

TRAINER
+-- Dashboard
+-- My Batches

```

+-- My Students
+-- Attendance
+-- Topics
+-- Assignments
+-- Exams
+-- Performance
+-- Reports
= 9

```

HR

```

+-- Dashboard
+-- Interview Ready
+-- Interviews
+-- Results
+-- Placement
= 5

```

STUDENT

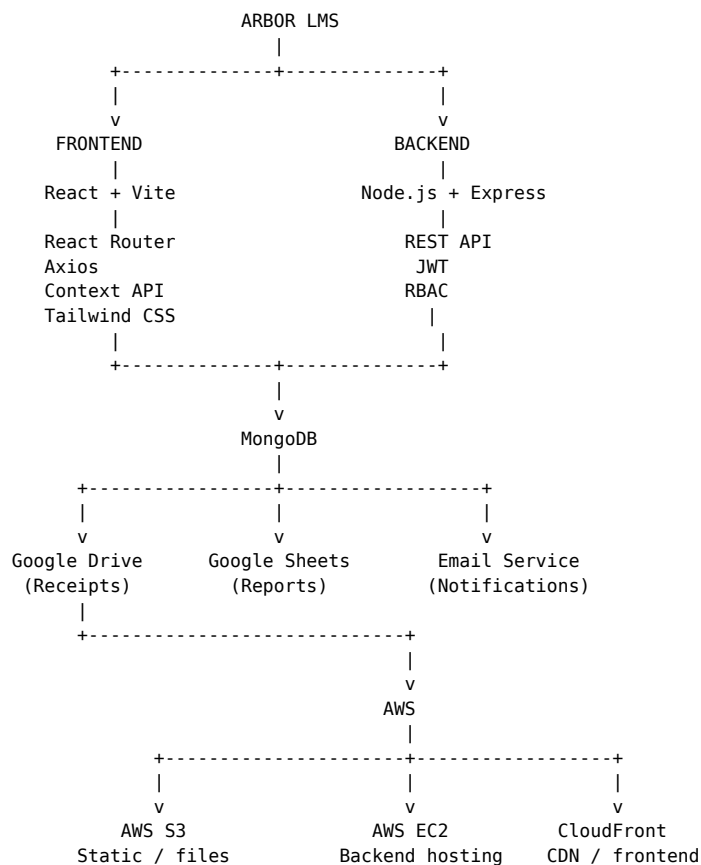
```

+-- Dashboard
+-- Profile
+-- My Course
+-- Topics
+-- Attendance
+-- Assignments
+-- Exams
+-- Performance
+-- Progress
+-- Placement
= 10

```

Approx. 50 logical screens, with reusable components and modals.

24. FINAL TECHNOLOGY ARCHITECTURE



25. RECOMMENDED PROJECT IMPLEMENTATION ORDER

PHASE 1

- > Create React + Vite frontend
- > Create Node + Express backend
- > Connect MongoDB
- > Create Login

- > Implement JWT authentication
- > Implement role-based routing

PHASE 2

- > Admin dashboard
- > User management
- > Course management
- > Batch management
- > Trainer management

PHASE 3

- > Sales admission
- > Fee status
- > Receipt upload
- > Student ID generation
- > Student account creation

PHASE 4

- > TL dashboard
- > Batch assignment
- > Trainer assignment
- > Student transfer
- > Student progress

PHASE 5

- > Trainer dashboard
- > Attendance
- > Topics
- > Assignments
- > Exams
- > Performance
- > Reports

PHASE 6

- > Student dashboard
- > Course content
- > Progress bars
- > Attendance
- > Assignments
- > Exams

PHASE 7

- > HR dashboard
- > Interview readiness
- > Interview scheduling
- > Results
- > Placement

PHASE 8

- > Google Drive API
- > Google Sheets API
- > Email notifications

PHASE 9

- > AWS S3
- > AWS EC2
- > AWS CloudFront
- > Domain / HTTPS later

PHASE 10

- > Testing
- > Security review
- > Role/permission testing
- > Production deployment
- > GitHub documentation

26. BEST STACK FOR YOUR PROJECT

FRONTEND

React + Vite
 React Router
 Axios
 Context API
 Tailwind CSS

BACKEND

Node.js
 Express.js
 JWT

bcrypt
Multer
Mongoose

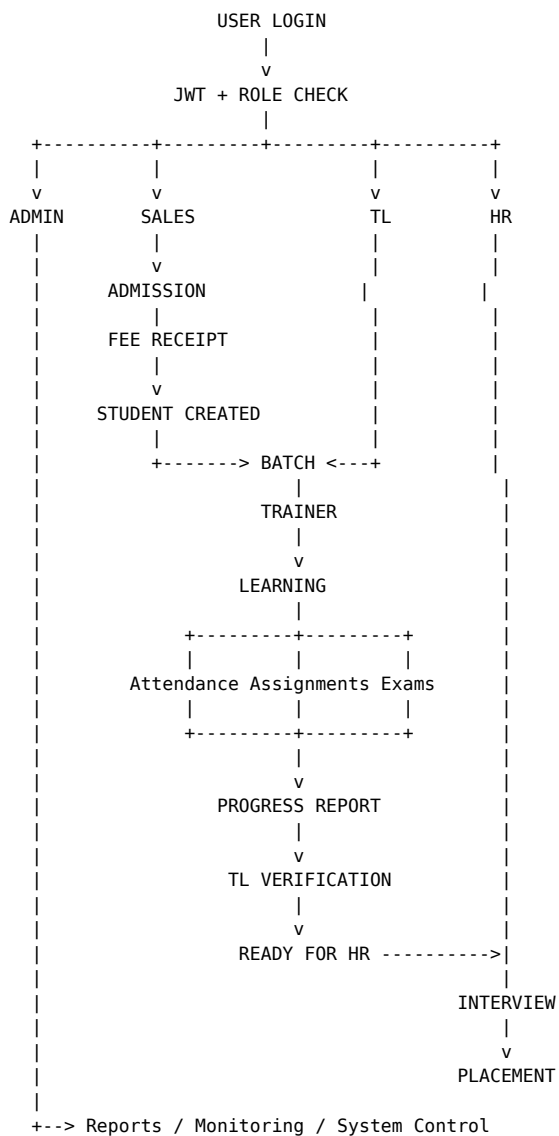
DATABASE
MongoDB

GOOGLE INTEGRATION
Google Drive API
Google Sheets API
Email Service

AWS
AWS S3
AWS EC2
AWS CloudFront

Optional later
Route 53
AWS Certificate Manager
CloudWatch

27. FINAL END-TO-END DATA FLOW



28. PROJECT GOAL

The finished Arbor Academy LMS should provide one central system where:

Sales
-> handles admissions and fee receipts

TL

-> manages batches, trainers and student allocation

Trainer / Sir

-> handles teaching, attendance, assignments, exams and progress

Student

-> learns, submits work, gives exams and tracks progress

HR

-> handles interview and placement workflow

Admin

-> controls the complete academy

Google Drive

-> stores receipts and documents

Google Sheets

-> provides operational/reporting views

MongoDB

-> stores application data

AWS

-> hosts and scales the application

Implementation Notes

1. Use React + Vite together; Vite is the build/development tool and React is the UI library. 2. Keep business logic in the Node.js backend, not in the React client. 3. Enforce permissions on the backend even if the frontend hides menu items. 4. Hash passwords with bcrypt and use short-lived access tokens plus a secure refresh-token strategy for production. 5. Store file metadata/URLs in MongoDB while the actual receipt files live in Google Drive or S3. 6. Use unique IDs for students and unique batch IDs for every course/batch combination. 7. Keep audit logs for important actions such as fee receipt upload, batch transfer, trainer assignment, and interview status changes.

Suggested first milestone: build Login → RBAC → Admin/TL/Trainer/Student dashboards → Course/Batch/Student models → Attendance/Progress. Add Sales/Fees/Drive and HR/Placement after the core LMS works.